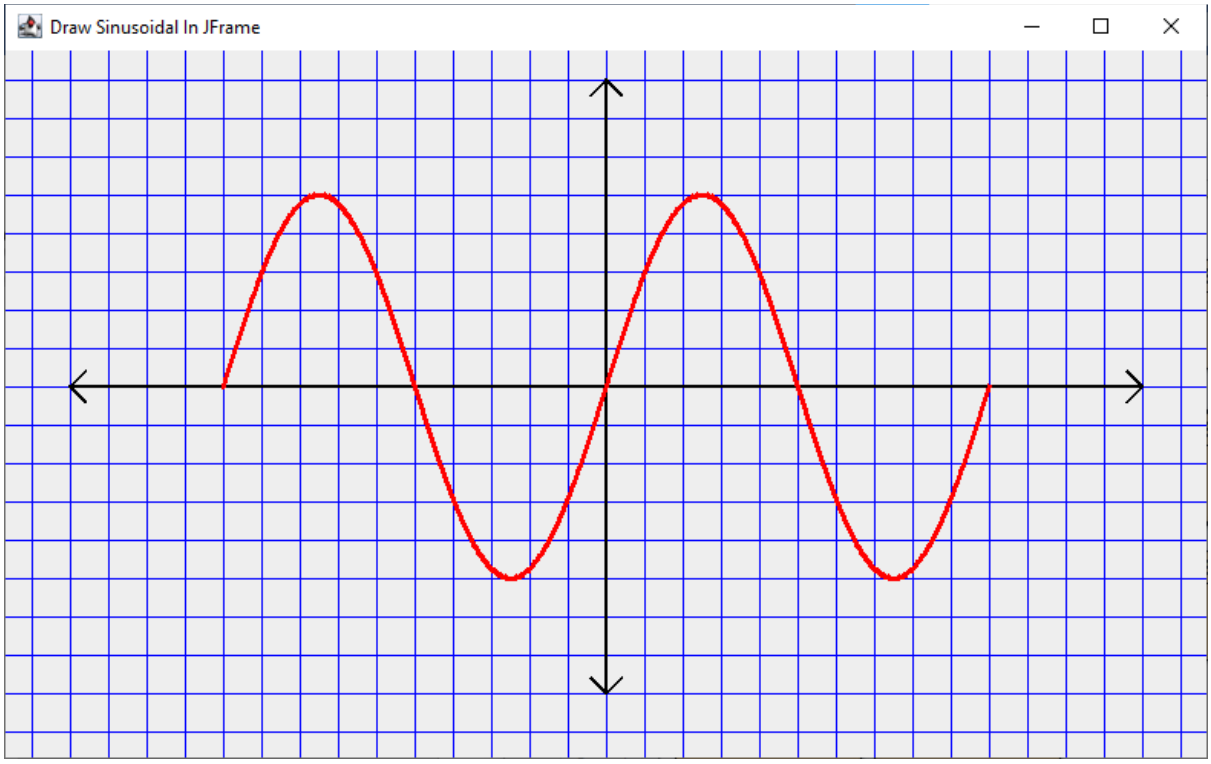
Nama : Abdul Rozak A
NIM : 221344033
Kelas : 2B – TNK

Rangkuman 06 Apk. Komputer dan Basis Data




```
import javax.swing.JFrame;
import java.awt.Graphics;
import java.awt.Color;
import java.awt.*;
import java.awt.geom.*;

public class L06_03 extends JFrame
{
public L06_03()
{
 super("Draw Sinusoidal In JFrame"); //Set JFrame title
 setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
 setSize(800,500);
 setVisible(true);
}

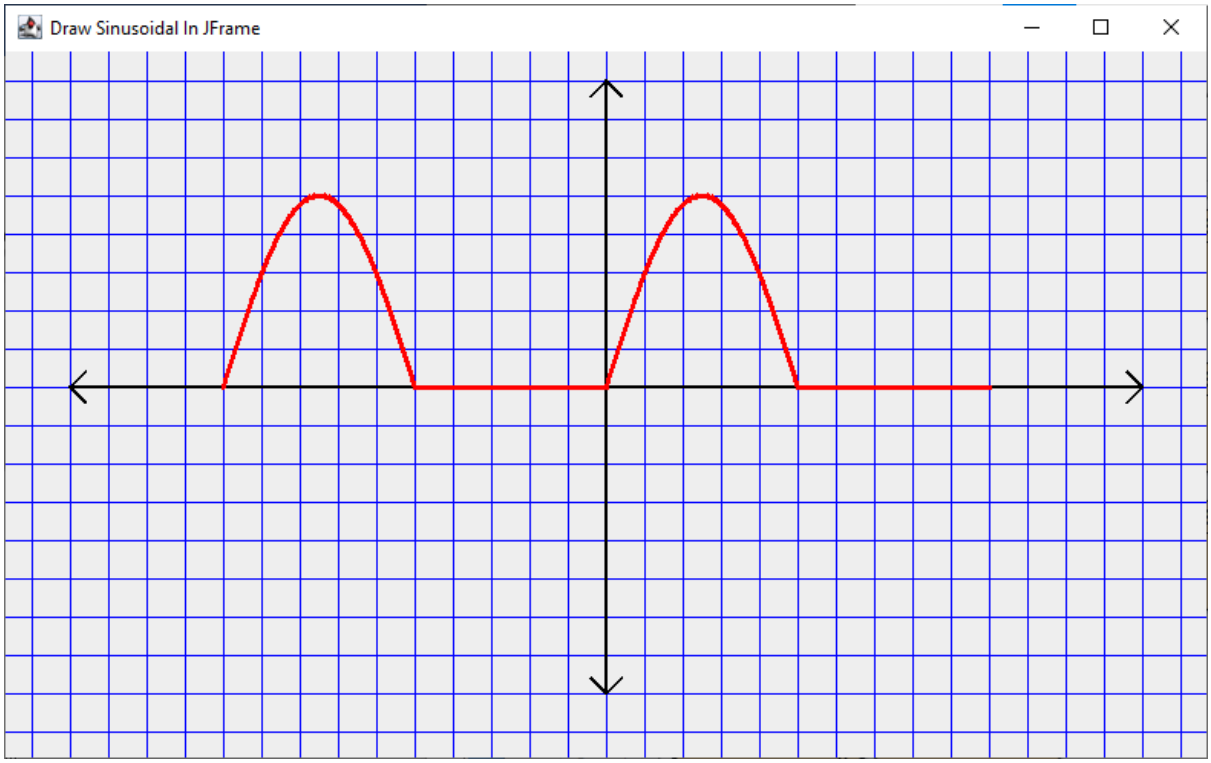
public void paint(Graphics g)
{
 super.paint(g);
 Graphics2D g2 = (Graphics2D)g;
 int a=0, b=0;
  double radian=0, radian2=0, sinus=0, sinus2=0;
```

```
 double cosinus, cosinus2;
     g.setColor(Color.blue);
     for(int i=25;i<800;i+=25)
     g.drawLine(i,0,i,500);
     for(int i=25;i<500;i+=25)
     g.drawLine(0,i,800,i);
     g.setColor(Color.black);
     g2.setStroke(new BasicStroke(2));
     g.drawLine(400,50,400,450);
     g.drawLine(50,250,750,250);
     for(int i=0;i<2;i++){
        g2.rotate(Math.toRadians(180),400,250);
        g.drawLine(400,50,390,60);
        g.drawLine(400,50,410,60);
        g.drawLine(50,250,60,240);
        g.drawLine(50,250,60,260);
     }
     //====================
     g2.setStroke(new BasicStroke(3));
     for(int i=0; i<2*360/1.44; i++ ){
       a=i; b=a+1;
       //default 1 perioda = 360px karena dilakukan 360 kali loop (i) ~ 14.4 grid
       //supaya default menjadi 1 perioda = 10 grid ==> 14.4/10 = 1.44
       //sampling (i) menjadi dibagi 1.44 dan 1.44 dikali radian
       radian = Math.toRadians(i);
       sinus = (Math.sin(1.44*radian*1));//radian dikali frekuensi

       radian2 = Math.toRadians(i+1);
       sinus2 = (Math.sin(1.44*radian2*1));//radian dikali frekuensi

     for(int j=0; j<=1; j++){
       g2.setColor(Color.red);
       Line2D line=new Line2D.Double((150+a),250-(125*sinus),(150+b),250-(125*sinus2));
       //yg disebelah a sama b titik mulai, yang disebelah sinus ampl,250 titik tengah sb-y
       g2.draw(line);
     }
   }
}

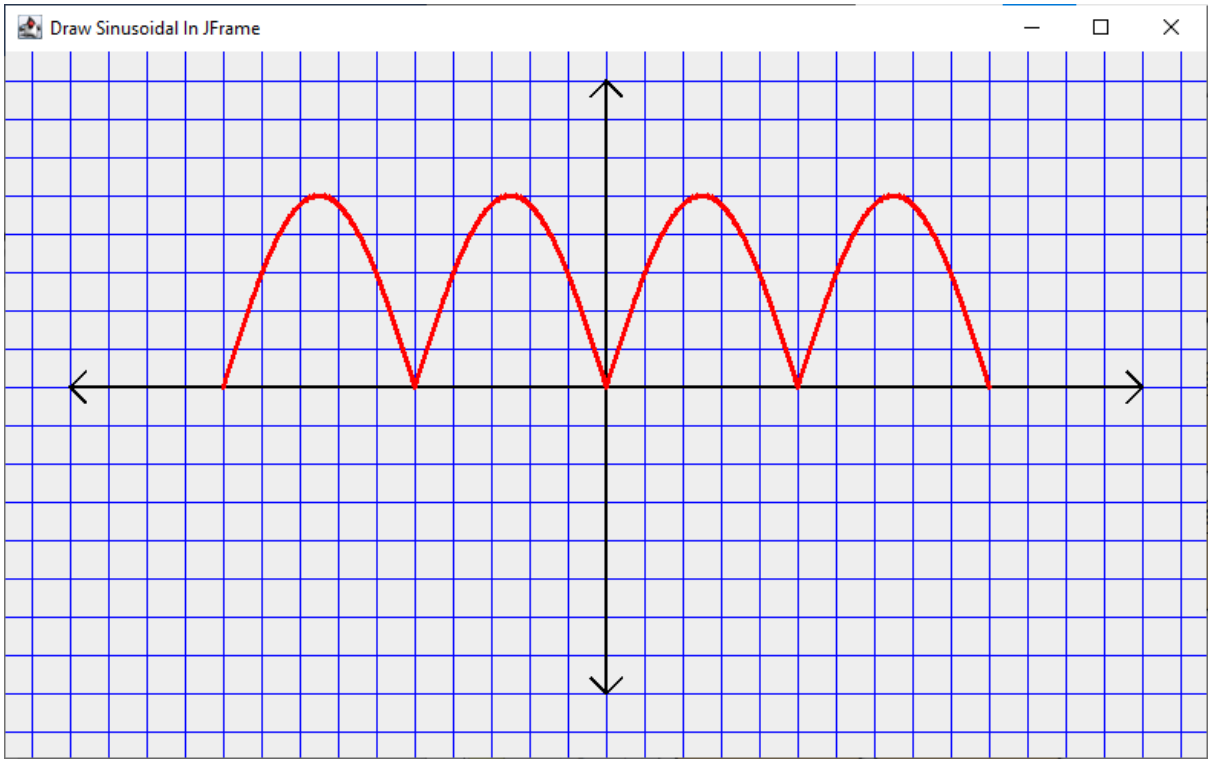
public static void main(String[]args)
{
 L06_03 gambar = new L06_03();
}
}
```




```
for(int i=0; i<2*360/1.44; i++ ){
        a=i; b=a+1;
        //default 1 perioda = 360px karena dilakukan 360 kali loop (i) ~ 14.4 grid
        //supaya default menjadi 1 perioda = 10 grid ==> 14.4/10 = 1.44
        //sampling (i) menjadi dibagi 1.44 dan 1.44 dikali radian
        radian = Math.toRadians(i);
        sinus = (Math.sin(1.44*radian*1));//radian dikali frekuensi
        if(sinus<0)sinus=0;//kondisi untuk halfwave

        radian2 = Math.toRadians(i+1);
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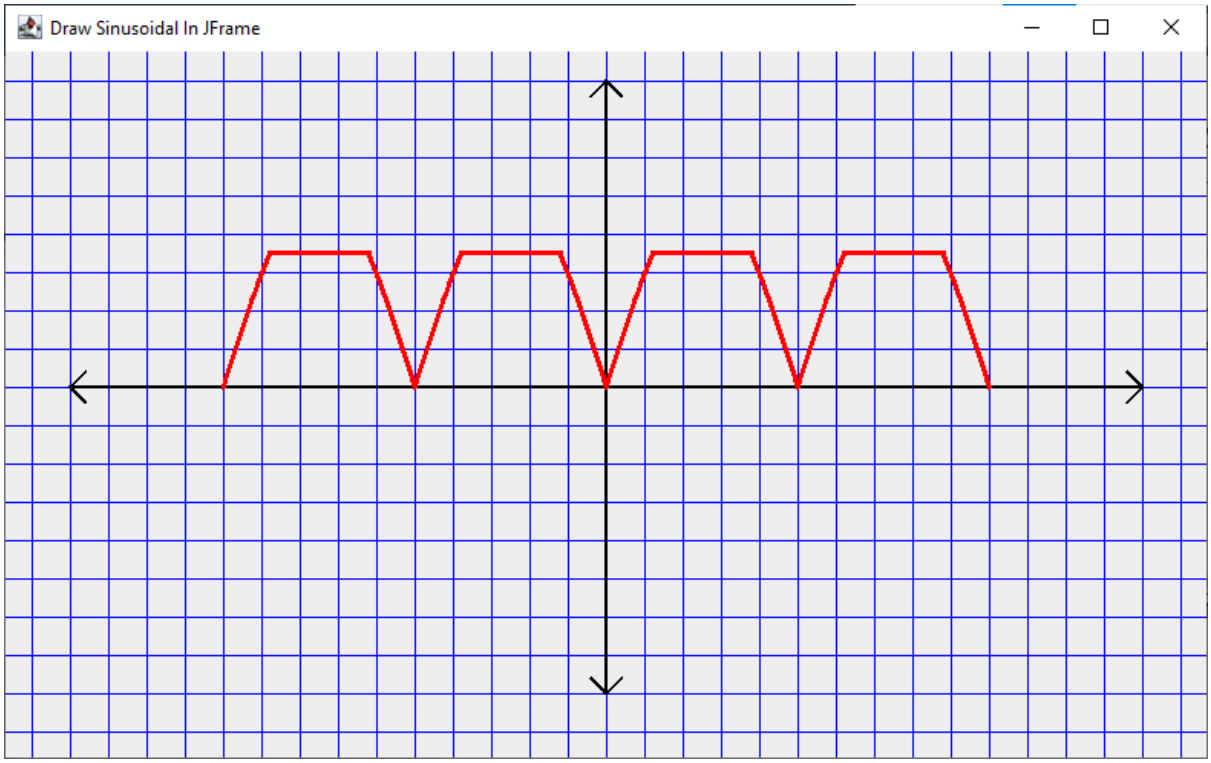
     for(int j=0; j<=1; j++){
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        Line2D line=new Line2D.Double((150+a),250-(125*sinus),(150+b),250-(125*sinus2));
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        g2.draw(line);
     }
  }
```




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        if(sinus<0)sinus=-sinus;//kondisi untuk fullwave

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        g2.draw(line);
     }
  }
```




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        radian2 = Math.toRadians(i+1);
        sinus2 = (Math.sin(1.44*radian2*1));//radian dikali frekuensi
        if(sinus2<0)sinus2=-sinus2;//kondisi untuk fullwave

        if(sinus>0.7||sinus2>0.7){//memotong sinyal
          sinus=0.7;
          sinus2=0.7;
        }

    for(int j=0; j<=1; j++){
      g2.setColor(Color.red);
      Line2D line=new Line2D.Double((150+a),250-(125*sinus),(150+b),250-(125*sinus2));
      //yg disebelah a sama b titik mulai, yang disebelah sinus ampl,250 titik tengah sb-y
      g2.draw(line);
    }
  }
```